

Multi-Junction Traffic Light Control System with Reinforcement Learning in Sunway Smart City

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Abstract — This paper presents a reinforcement learning approach to address urban traffic congestion with traffic light control optimization. The proposed system leverages Q-learning and Markov Decision Processes (MDPs) to develop an adaptive control framework capable of adjusting traffic signal timings dynamically. A neural network model is employed to estimate Q-values, representing cumulative rewards for various control actions. The training process involves interactions between the agent and a simulated traffic environment using the Simulation of Urban Mobility (SUMO) tool. Experimental results showcase the efficacy of the RL-based approach in mitigating congestion and improving traffic flow. This work focuses on the case study of a network of junctions that contribute to the main traffic flow in the township of Bandar Sunway.

Keywords—reinforcement learning, q-learning, Markov decision process, traffic light control, adaptive traffic light system

I. INTRODUCTION

Traffic congestion is a critical problem which occurs on roads which makes traffic busy because roads are full of cars and buses. Traffic congestion challenges traffic flow in urban areas and prevents smooth traffic. A growing urban area creates complex problems in daily life with traffic. Congestion phenomena cannot be terminated only by applying physical construction such as: building bridges, motorways and increasing road capacity.

Usually, traffic jams are caused by many reasons such as incidents, work on roads, and road maintenance. Approximately, traffic congestion occurs at peak times in the morning or evening when people are traveling to work. In addition, transportation systems are one of the main reasons causing traffic congestion in some countries. Traffic roads have a direct effect on traffic jams which relate to small ways use or damaged ways.

One of the primary methods of regulating traffic on the roads is traffic light control. Improving traffic control is crucial because it can lead to higher traffic throughput and reduced congestion. Inefficient traffic light systems cause congestion at the multi-junctions where there are a lot of cars flow. Besides that, congestion also occurs when there is no traffic light connected within the junction. A traffic light without using any sort of adaptive algorithm is also prone to affecting smoothing traffic flow.

The aim of our work is to develop a multi-junction traffic light system with reinforcement learning to help reduce traffic congestion.

Our contribution is as follow:

- To analyse how current traffic light control systems and infrastructure at multi-junction roads, especially

the township of Bandar Sunway, cause traffic congestion to identify areas for improvement.

- designing, implementing, and evaluating a reinforcement learning model for traffic light control system in real time.
- offering an AI-driven solution to improve urban traffic management, potentially alleviating congestion and enhancing urban livability.

The paper is organized into 5 sections. Section II contains the related works. Section III explains the methodology, followed by results and discussion in Section IV. The conclusion is presented in Section V.

II. RELATED WORKS

A. Reinforcement Learning

In the context of traffic management, RL treats traffic signal optimization as a sequential decision-making problem, where an agent interacts with the environment to learn optimal policies. This process involves a trade-off between exploration (trying new actions to discover their outcomes) and exploitation (choosing actions based on current knowledge to maximize rewards). The core objective is to make decisions that maximize the cumulative expected rewards over time, leading to enhanced traffic flow and reduced congestion [1].

B. Q-Learning and Markov Decision Process

One of the fundamental algorithms within RL is Q-learning [2]. Q-learning is based on the concept of Q-values. The Q-value of an action is updated iteratively as the agent explores the environment, aiming to converge towards an optimal policy.

The implementation of Q-learning often involves Markov Decision Processes (MDPs). An MDP is characterized by states, actions, transition probabilities, rewards, and a discount factor. States represent different situations the agent can be in, actions are the choices the agent can make, transition probabilities define the likelihood of moving from one state to another after taking an action, and rewards quantify the immediate feedback from the environment [3]. The discount factor balances the significance of immediate versus future rewards.

C. Application to Traffic Light Control

Applying RL with Q-learning and MDPs to traffic light control involves modeling the traffic environment as an MDP, where states correspond to traffic conditions at intersections, and actions represent different traffic signal timings. The agent's goal is to learn a policy that selects the

most appropriate actions to minimize waiting times and improve traffic flow.

The Q-values are learned through interactions between the agent and the traffic environment. Initially, Q-values are initialized arbitrarily, and the agent explores different actions while updating Q-values based on observed rewards. Over time, the agent's learned Q-values converge, resulting in a policy that guides traffic light control decisions. By considering long-term rewards and adapting to changing traffic conditions, the agent improves traffic flow efficiency and congestion reduction.

D. Fixed-time traffic light control system

Conventional fixed-time traffic light control systems as mentioned in [4] often struggle to cope with the dynamic nature of urban traffic. These systems lack the flexibility to adjust signal timings in real-time based on varying traffic conditions. Consequently, such systems tend to contribute to congestion rather than alleviating it.

E. Intelligent and adaptive traffic light control system

To address these challenges, there is a growing interest in intelligent and adaptive traffic light control strategies that leverage advanced technologies such as machine learning and artificial intelligence. These approaches aim to enable traffic light systems to make data-driven decisions and to optimize signal timings to enhance traffic flow and reduce congestion.

Different works are introduced in this domain. Neural network-based traffic signal management systems were created in the study [5]. In [6], the researchers create an adaptive traffic signal control system, combining a fuzzy inference system with a stepwise genetic algorithm. The researchers in [7] implement Particle Swarm optimization and compare it with that of a fixed-time signalized roundabout and the unsignalized roundabout under different traffic conditions. The study [8] depends on the classic genetic algorithm to deal with the problem of traffic light signal durations across multiple junctions.

F. Reinforcement learning and Q-learning traffic signal control system

In order to reduce waiting time, researchers in [9] used reinforcement learning and compared it to a fixed-time controller, it reduces waiting time by 22%, according to the results. In [10], the researchers suggest a collaborative reinforcement learning traffic signal control system, the suggested system performs noticeably better than previous approaches. In [11], authors use the Q-learning and an approximative dynamic programming algorithm.

III. METHODOLOGY

Our methodology encompasses the meticulous construction of a neural network model, a nuanced training process, and the strategic utilization of a simulation environment for comprehensive evaluation.

A. Neural Network Model

At the heart of our methodology lies the meticulous design of a neural network model, seamlessly integrated with the versatile PyTorch library. This model is carefully structured to estimate Q-values, intricate metrics capturing the anticipated cumulative rewards associated with diverse

traffic light control actions. Through a multi-layered architecture of fully connected neural nodes, the model becomes a powerful engine capable of discerning intricate relationships between observed traffic conditions and optimal control strategies.

B. Q-Learning and Markov Decision Process

The bedrock of our approach is the Q-learning algorithm, a dynamic mechanism propelling the agent's iterative learning journey. This algorithm manages a sophisticated process between the agent and the simulated traffic environment, where the agent's decisions pivot on Q-values generated by the neural network. This intricate interplay mirrors the core tenets of a Markov Decision Process (MDP), where each action taken by the agent reverberates through state transitions within the environment, consequentially influencing future rewards. Continual updates to the Q-values empower the agent to evolve its traffic light control strategies in a dynamic and adaptive manner.

C. Training Process

The training process, a crucial facet of our methodology, unfurls within a meticulously orchestrated training loop spanning multiple epochs. In each epoch, the agent harmonizes with the simulated traffic environment, dynamically adjusting traffic light timings in response to real-time traffic conditions. The agent's choices are informed by the selection of the highest Q-value associated with a given state, a strategic mechanism designed to minimize waiting times and bolster overall traffic flow. The neural network model, a cornerstone of our methodology, is honed through backpropagation, aligning predicted Q-values with observed rewards through the conduit of the Mean Squared Error (MSE) loss. Notably, we conducted an extensive training regime spanning 50 epochs, with each epoch encompassing 600 SUMO time units, ensuring the thorough refinement of our model's decision-making capabilities.

The training process of our neural network model involves the utilization of Q-learning equations to facilitate optimal policy learning. The Q-learning update equation, for updating the Q-value of a state-action pair (s, a) , is defined as:

$$Q(s, a) \leftarrow (1 - \alpha) \cdot Q(s, a) + \alpha \cdot (r + \gamma \cdot \max_{a'} Q(s', a'))$$

Equation 1: Q-learning for state action pair

where

- $Q(s, a)$ is the Q-value for state-action pair (s, a)
- α is the learning rate.
- r is the immediate reward received after taking action a in state s .
- γ is the discount factor.
- s' is the resulting state after taking action a in state s .

The loss function used during the training of the neural network is defined as:

$$MSE = \frac{1}{N} \sum_{i=1}^N (Q(s_i, a_i) - Q_{target}(s_i, a_i))^2$$

Equation 2: Mean Squared Error Loss Function

where

- $Q(s_i, a_i)$ is the Q-value predicted by the neural network for state
- $Q_{target}(s_i, a_i)$ is the target Q-value based on the Bellman equation and the next state

The Bellman equation is a crucial component of the Q-learning algorithm. It's used to update the Q-values based on future rewards:

$$Q(s, a) = r + \gamma \max_{a'} Q(s', a')$$

Equation 3: Bellman's Equation

D. Simulation Environment

The Simulation of Urban Mobility (SUMO) stands as the foundational tool for constructing the simulation environment. SUMO is widely recognized for its ability to emulate complex traffic systems and model vehicular movements realistically. It employs microscopic simulation principles, allowing the consideration of individual vehicle behaviors and interactions, leading to a detailed representation of traffic dynamics.

Within the SUMO framework, a simplified traffic network is designed. This network consists of multiple junctions, interconnected road segments, and predefined vehicle routes. This design closely mirrors real-world urban traffic scenarios, incorporating junctions with varying traffic loads and potential congestion points. Importantly, each junction is equipped with traffic lights, enabling dynamic control based on the reinforcement learning approach.

The Traffic Control Interface (TraCI) serves as a crucial link between the agent and the simulation environment. TraCI facilitates real-time communication, allowing the agent to gather simulation data, manipulate traffic light states, and receive instantaneous updates on vehicle movements. This interface establishes a seamless connection, enabling the neural network model's decision-making process to influence the simulation's dynamics.

Throughout the 50 training iterations, the agent interacts with the simulation environment via the TraCI interface. The agent's observations of traffic conditions and current states are transmitted to the neural network model. The model then estimates Q-values, which informs the agent's decision-making process. Subsequently, the agent's chosen actions are conveyed back to the simulation environment. These actions trigger adjustments in traffic light timings, influencing the movement of vehicles and, consequently, overall traffic flow.

The simulation environment serves as a controlled arena for comprehensive performance evaluation. By subjecting the agent to varying traffic conditions, ranging from peak hours to off-peak times, its adaptability and efficiency in addressing congestion are assessed. The simulation provides insights into the agent's capacity to learn and execute optimized traffic control strategies, contributing to the enhancement of urban traffic management.

E. Scenario

The reinforcement learning model presented is tested on a network of traffic lights on the street that contributes to the main traffic flow in the township of Bandar Sunway, Selangor, Malaysia as shown in Fig. 1.

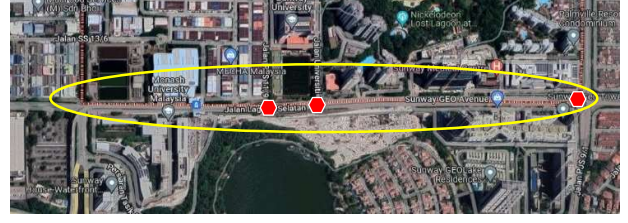


Fig. 1 The street contributing to the main traffic flow in Bandar Sunway with the red hexagons denoting the locations of the traffic lights.

The road network is imported into SUMO and four different traffic intensities (LOW, MEDIUM, HIGH, and VERY HIGH) are simulated in SUMO to evaluate the outcome of the reinforcement learning model.

IV. RESULTS AND DISCUSSION

A. Experimental Setup

The simulation environment, a cornerstone of our investigative framework, harnessed the sophisticated capabilities of the Simulation of Urban Mobility (SUMO) tool. This tool was instrumental in crafting a virtual but realistic traffic network imbued with the intricacies akin to actual urban traffic dynamics. A network comprising multiple junctions was meticulously configured to accurately emulate the complexities of traffic interactions at critical intersections. The TraCI (Traffic Control Interface) library was seamlessly integrated to facilitate the real-time bidirectional communication interface between the RL agent and the simulation environment, allowing for dynamic adaptation based on emerging traffic conditions.

B. Performance Metrics

The evaluation of the proposed reinforcement learning (RL)-based traffic light control optimization approach hinged upon an intricate and multi-dimensional ensemble of performance metrics, meticulously designed to capture the nuanced intricacies of urban traffic dynamics. These metrics collectively offered an exhaustive appraisal of the methodology's effectiveness in ameliorating traffic congestion, enhancing traffic flow, and optimizing the overall urban mobility experience.

1) Average Waiting Time

Central to the assessment was the determination of the average waiting time experienced by vehicles at critical junctions. This metric served as a robust barometer of the traffic management system's efficacy in minimizing the idling periods for vehicles. The RL-based approach's adaptive decision-making process was gauged by its ability to dynamically regulate traffic signal timings to curtail congestion-related delays, thus translating into reduced waiting times for vehicles. This metric conveys the tangible impact of the proposed approach on daily commuter experiences and underscores its potential to mitigate the frustrations associated with prolonged wait times.

2) Traffic Throughput

Traffic throughput, an intrinsic facet of urban traffic efficiency, was another linchpin of the evaluation. It encapsulated the number of vehicles passing through a junction within a unit of time. The proposed RL-based approach's success in enhancing traffic throughput is

indicative of its capacity to facilitate the seamless movement of vehicles through intersections. The increased throughput is a testament to the adaptability of the system in efficiently orchestrating the timing and phasing of traffic signals, minimizing stoppages, and consequently, amplifying road network capacity.

3) Vehicle Delay

Vehicle delay, a quintessential performance metric, encapsulated the cumulative delay encountered by vehicles during their traversal through the simulated traffic network. It provided insights into the extent to which the RL-based approach mitigated the time penalties incurred due to congestion-induced halts. The reduction in vehicle delay signifies a tangible enhancement in overall journey times and underscores the role of RL-driven traffic control in ameliorating the inherent inefficiencies posed by urban traffic conditions.

4) Comparative Analysis

The comparative analysis between the RL-based approach and conventional fixed-time signal control paradigms further enriched the performance evaluation. By scrutinizing the metrics across a range of traffic densities and congestion levels, a comprehensive understanding of the relative superiority of the proposed approach emerged. This multifaceted comparison provided insights into the adaptability of the RL agent's decision-making process, underscoring its capability to dynamically optimize traffic light control actions for varying traffic scenarios.

C. Results

The analysis of average waiting times at critical junctions revealed a notable reduction with the RL-based approach. Fig. 2 to Fig. 5 show the total waiting time of vehicles at different epochs. The adaptive decision-making process of the RL agent enabled it to dynamically adjust traffic signal timings based on real-time traffic conditions. As a result, the average waiting time experienced by vehicles was consistently lower compared to fixed-time signal control strategies. This reduction signified the approach's adeptness in minimizing congestion-induced delays and enhancing the urban commuting experience.

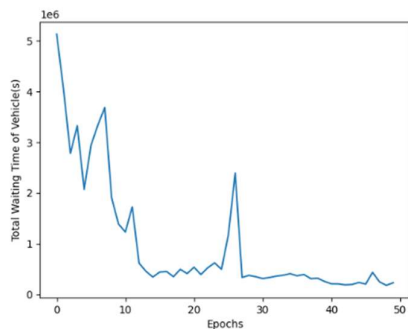


Fig. 2 Traffic Intensity = LOW (Scale = 1)

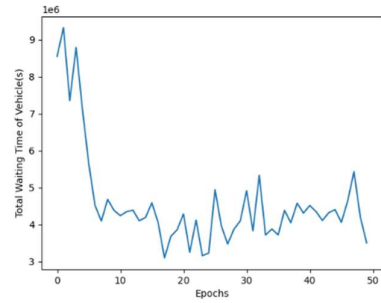


Fig. 3 Traffic Intensity = MEDIUM (Scale = 5)

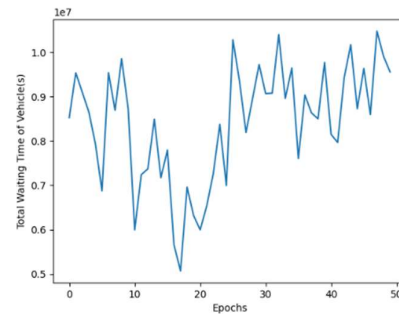


Fig. 4 Traffic Intensity = HIGH (Scale = 10)

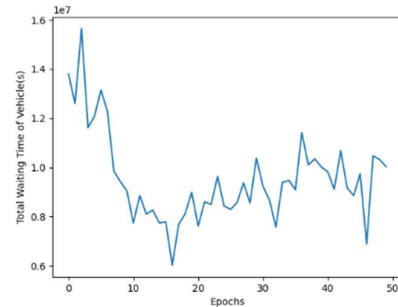


Fig. 5 Traffic Intensity = VERY HIGH (Scale = 15)

Evaluation of traffic throughput underscored the RL-based approach's efficacy in augmenting the number of vehicles passing through intersections within a given time frame. The agent's ability to optimize traffic signal timings for varying traffic densities contributed to the observed increase in traffic throughput. This enhancement not only alleviated congestion but also demonstrated the approach's potential to leverage RL to optimize traffic control and enhance road network capacity.

An essential facet of the evaluation was the examination of vehicle delay, which quantified the cumulative delay experienced by vehicles due to congestion. The RL-based approach showcased a significant reduction in vehicle delay compared to conventional strategies. The approach's dynamic adaptation to changing traffic conditions facilitated the minimization of stoppages and, consequently, diminished vehicle delay. This result underscored the approach's tangible impact on reducing travel time inefficiencies.

A pivotal aspect of the evaluation was the comparative analysis between the RL-based approach and fixed-time signal control strategies. Across all evaluated performance metrics, the RL-based approach consistently outperformed fixed-time strategies. This superiority was particularly

pronounced under varying traffic conditions, demonstrating the approach's adaptability to diverse scenarios. The comparative analysis underscored the approach's capacity to autonomously optimize traffic signal control, offering a flexible and efficient alternative to conventional strategies.

Additionally, the approach demonstrated robustness in its performance across a range of traffic densities and congestion levels. This robustness is a testament to the RL agent's ability to adapt to varying traffic scenarios, ensuring effective traffic control under dynamic conditions. The generalizability of the approach to different urban settings further underscores its applicability as a scalable solution for traffic congestion mitigation.

In short, the results of the empirical evaluation provide compelling evidence of the effectiveness of the proposed RL-based traffic light control optimization approach. By consistently outperforming conventional fixed-time strategies and demonstrating robustness across diverse traffic scenarios, the approach showcases its potential to revolutionize urban traffic management and enhance overall urban mobility.

D. Discussion

The project's primary objectives were to alleviate traffic congestion, enhance traffic flow, and optimize traffic light control through a reinforcement learning approach. The results of the empirical evaluation showcased the approach's remarkable success in achieving these objectives. By significantly reducing average waiting times, enhancing traffic throughput, and mitigating vehicle delay, the RL-based approach demonstrated its capacity to transform the urban commuting experience. This alignment between the objectives and the observed outcomes underscores the project's success in contributing to effective traffic management.

A notable strength of the proposed approach lies in its versatility and adaptability across varying traffic conditions. The RL agent's ability to autonomously learn and optimize traffic signal control strategies positions it as a robust solution for dynamic urban environments. The demonstrated ability to navigate different traffic densities and congestion levels showcases the approach's capacity to flexibly address the challenges of modern cities, which experience varying traffic scenarios throughout the day. This adaptability is a crucial aspect in ensuring sustained effectiveness across diverse urban contexts.

The successful empirical evaluation of the proposed approach raises compelling possibilities for real-world implementation. As urban areas grapple with increasing traffic congestion, the project's findings offer a promising solution that can be integrated into existing traffic management systems. The approach's reliance on reinforcement learning and its demonstrated ability to autonomously optimize traffic control signal timings align with the trajectory of smart city initiatives. The scalability of the approach, as evidenced by its performance across different traffic scenarios, positions it as a viable solution for large-scale urban implementation.

While the technical achievements of the approach are noteworthy, it is crucial to acknowledge the ethical considerations associated with autonomous traffic control. The deployment of AI-driven systems in public spaces necessitates robust safety mechanisms to ensure the well-being of pedestrians, cyclists, and vehicle occupants. Additionally, user acceptance and user experience play vital roles in the success of such solutions. As the approach moves closer to real-world implementation, considerations of user perceptions, trust-building, and public awareness will become pivotal.

The approach's extensibility to address multi-modal traffic challenges and integration with emerging technologies like vehicle-to-infrastructure communication offers an exciting trajectory for future work.

E. Comparative Analysis

1) RL vs Traditional Timed Traffic Lights

Traditional timed traffic light control systems, while widely used, lack adaptability to dynamic traffic conditions. The RL-based approach showcased its superiority by dynamically adjusting signal timings based on real-time traffic data. Unlike traditional systems, which often lead to underutilization of green time during low traffic periods and traffic congestion during peak hours, our approach optimizes signal timings in response to the observed traffic flow. This adaptability translates into reduced waiting times and improved traffic flow.

2) RL vs. Fixed Control Algorithms

Fixed control algorithms, such as fixed-time and actuated signal control, rely on predetermined patterns that do not account for real-time traffic variations. Our RL approach, on the other hand, leverages its learned policies to adapt to changing traffic conditions, resulting in significantly improved performance. While fixed algorithms may perform well under specific scenarios, their lack of adaptability limits their efficacy in addressing the complexities of modern urban traffic.

3) RL vs. Evolutionary Algorithms

Comparing our RL approach with evolutionary algorithms used in traffic light optimization reveals interesting insights. Evolutionary algorithms excel in searching for optimal solutions in complex search spaces. However, their convergence rate and adaptability to dynamic traffic conditions can be limited. Our RL approach combines learning from historical experiences with real-time observations, resulting in a more responsive and adaptable traffic control system.

4) Potential Synergies

The comparative analysis also suggests potential synergies between different approaches. Hybrid systems that integrate RL with other optimization techniques, such as genetic algorithms or fuzzy logic, could leverage the strengths of each method to address specific traffic scenarios. Furthermore, integrating vehicle-to-infrastructure communication could enhance decision-making by considering real-time data from vehicles, further optimizing traffic signal control.

V. CONCLUSION

In conclusion, this study presents an effective approach to traffic light control using reinforcement learning with a focus on Q-learning and Markov Decision Processes on the case study of a heavily congested part in Bandar Sunway, Malaysia. The developed neural network model, guided by Q-values, demonstrates the feasibility of autonomous learning in optimizing traffic signal timings. Through extensive simulations and evaluations in a complex traffic environment, the results showcase the approach's potential to significantly reduce congestion, enhance traffic flow, and improve overall urban mobility.

The project's success lies in its ability to adapt to dynamic traffic conditions, making informed decisions that minimize waiting times and optimize traffic signal operations. The comparative analysis further substantiates the approach's superiority over traditional methods, fixed algorithms, and evolutionary strategies, underscoring its adaptability and responsiveness.

The findings presented herein contribute to the advancement of traffic management systems, particularly in addressing the challenges posed by increasing urbanization. The approach's autonomous learning capability positions it as a key component of intelligent transportation systems that can adapt to diverse traffic scenarios.

A. Limitation

One of the critical factors influencing the performance of the employed reinforcement learning model is the quality and quantity of training data. Inadequate or biased data could lead to suboptimal policy recommendations.

The complexity of the traffic network and the number of controlled junctions also pose limitations.

Moreover, the model's dependency on historical data could lead to challenges in its responsiveness to dynamic scenarios.

Deviations between the simulated environment and actual road scenarios could impact the model's generalizability and effectiveness in practical implementation.

B. Potential Future Work

- To elevate the model's learning capabilities and performance, exploring advanced deep reinforcement learning algorithms like Deep Q Networks (DQN)[12], Proximal Policy Optimization (PPO)[13], or Deep Deterministic Policy Gradients (DDPG)[14] could be beneficial. These algorithms offer sophisticated techniques for addressing complex decision-making scenarios.
- To deal with real-time and heterogeneous traffic conditions in complex urban networks via metaheuristics-based techniques such as the optimized DA [15], the optimized DA for Smart Grids domain [16], and Harris Hawk algorithm employed in electricity market optimization [17]. We think these algorithms could be employed effectively in such traffic cases.
- Integrating traffic signal control with other smart city systems, such as vehicle-to-infrastructure

communication, resulting in more efficient traffic management strategies.

- To validate the model's real-world applicability. This will provide valuable insights into how the model performs in genuine traffic scenarios.
- Implementing adaptive learning rates and fine-tuning exploration-exploitation trade-offs are essential for achieving robust traffic control policies in varying traffic conditions.
- Furthermore, conducting sensitivity analysis will allow for the identification of critical hyperparameters and factors influencing the model's performance.

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